

Scope: Gummel decoupled-Newton fallback for the steady-state driver

Date: 2026-06-22 **Status:** SCOPE + SCAFFOLD BUILT (M1+M2); M3/M4 OPEN **Owner area:** perovskite_sim/experiments/steady_state.py, physics/continuity.py, physics/interface_plane.py

0. Empirical update (2026-06-22) — scaffold built, basic Gummel STALLS

The M1+M2 scaffold is implemented (dormant, default-bit-identical): - `assemble_rhs(phi_frozen=...) + _residual_fn(phi_frozen=...)` — the frozen-phi carrier residual seam (`solver/mol.py`, `steady_state.py`). - `_gummel_point` — the decoupled outer loop (carrier Newton at frozen phi + `_qfl_poisson_relax` Poisson half + ln-space under-relaxation). Env-gated `SOLARLAB_SS_GUMMEL`; NOT wired into `run_jv_sweep_ss/solve_voc_ss` yet.

Measured on the deep-CBO point ($\Delta E_C = -1.0$, where the coupled Newton goes singular): the loop reduces the seed residual by ~13 orders ($4e13 \rightarrow O(1-1e9)$) but then **STALLS at a fixed point above the certification guard** — `res` ~3.6 with `iface_states`, ~2e9 without, and **identical at carrier_max_it 20 and 40**, i.e. a genuine *decoupled-Gummel stall*, not slow convergence. **This confirms the Section 9 primary risk: the basic decoupled scheme stalls at strong coupling.** So M1+M2 alone do NOT close the deep-CBO gaps. Closing them is the **M4 acceleration** work (the schedule risk): Anderson acceleration, a coupled-Newton polish that survives the singular Jacobian, or a transient-assisted Gummel. Until then the deep-CBO SS points stay documented as gapped (the transient covers that regime; SS ~ transient there). The `phi_frozen` seam is the reusable, bit-identical foundation M4 builds on.

M4 result (2026-06-22) — Anderson tried; deep-CBO confirmed INFEASIBLE and UNNECESSARY. CLOSED.

M4 was attempted and the leg is now **closed as won't-fix**. Two findings settle it:

(1) Anderson acceleration does not crack it. Anderson(`m=5`) was prototyped on the Gummel fixed-point map $G(z) = [\text{carrier-Newton-at-frozen-phi} \rightarrow \text{qfl-Poisson-relax}]$ (Type-II, f-difference least-squares mixing). Measured on four deep points ($\Delta E_C = -1.0 / -0.5$ at V in $\{0, 0.3, 0.7\}$, with and without `iface_states`): **STALL at res 3.0e9-6.5e9 after 120 iterations** — the same plateau as plain Gummel. The carrier half alone reduces 5 orders ($1.30e12 \rightarrow 4.76e7$ at it0) but the Poisson re-coupling pushes it back up and the outer map has a contraction factor ≥ 1 at this coupling strength; the Anderson extrapolation cannot fix a map whose fixed-point iteration diverges.

(2) Gummel is strictly WORSE than the shipped coupled Newton here, and the shipped driver already fails FAST. Best peak-density-relative residual ($\text{tol} = 1e-6$) at $\Delta E_C = -1.0$, $V = 0.3$:

solver	best residual	wall
coupled SS Newton (shipped <code>solve_steady_state</code>)	4.2e4	raises in 0.6 s
decoupled Gummel (M1+M2 scaffold)	1e9	stall
Anderson-accelerated Gummel (M4)	3e9	stall

All three are 10-15 orders above `tol`. The coupled Newton gets *closest* (4.2e4) and, critically, **raises a clean `SteadyStateError` in 0.6 s** (“line search stalled ... transient assists exhausted”) — it does **not** hang. So the deep-CBO SS root is genuinely pathological for *any* algebraic Newton (coupled or decoupled); the stiff modes only damp under a **long** transient (~8 s), which is exactly why the bounded SS transient-assist exhausts. A Gummel fallback adds a worse-conditioned path, not a better one.

Why this is also UNNECESSARY. CBO is a band-offset effect, not an interface-limited one — **SS approx transient** over the whole CBO sweep (verified in the flat-band region where both converge). The deep-CBO points therefore add **zero** scientific value as SS points: the transient curves already cover that regime in

every figure, and the full-FOM figures correctly carry transient-only there (the shipped driver's fast 0.6 s raise means the figure pipeline skips the point cleanly, no grind).

Disposition. M4 is closed won't-fix. The `_gummel_point` scaffold and the `phi_frozen` seam stay in-tree, **dormant + bit-identical** (env-gated `SOLARLAB_SS_GUMMEL`, not wired into any sweep), as the documented foundation if a *genuinely new* approach appears (pseudo-transient continuation with an unbounded dt ramp == the transient, so PTC reduces to "just run the transient"). The shipped solution for the deep tail is the existing transient driver, which is correct and fast. The net deliverable of this campaign is the **interface-channel calibration + Newton/LU reuse + stop_after_voc** (the operative regime — fast, accurate, complete on all 10 flat-band-operative sweeps); the deep-CBO tail is a documented engine boundary, not a gap to be closed with more solver machinery.

1. Motivation

The steady-state interface-states driver (`run_jv_sweep_ss(iface_states=True), solve_voc_ss`) is the configuration that recovers the SCAPS interface-doping directions (`Nd_ETL` rising, etc.). It has one structural weakness: the coupled damped-Newton (`solve_steady_state`) **fails to converge in collapsed-junction and extreme-band-offset regimes**, where it falls back to the certified transient point-fallback, which then grinds for minutes per point or fails.

Two concrete symptoms motivated this scope:

1. **Deep-CBO SS gaps (the trigger).** On the ETL/PVK conduction-band-offset sweep, the SS series converges only over the flat-band region ($\Delta E_C \geq -0.16$ eV). The deep-offset points ($\Delta E_C \leq -0.2$ eV, where V_{oc} collapses to ~ 0.33 - 0.83 V) **do not converge in SS** even at a 300 s cap — yet the **transient solves them in ~ 8 s** and finds a real $J=0$ crossing. So a crossing exists; the SS solver simply cannot reach it.
2. **The `Nd_ETL <= 1e10` xfail** (`tests/unit/experiments/test_steady_state.py` near-insulating gate) — see the critical caveat in Section 7; this is a *different* problem and is **not** fully solved by Gummel.

The de-spike + interface-channel calibration work (2026-06) made the operative regime fast and accurate; this scope is about the *hard tail* the coupled Newton can't reach.

2. Root cause

Every `assemble_rhs` call solves Poisson globally (`solver/mol.py:1693`), so `phi[i]` depends on `rho` at all nodes, and the Scharfetter-Gummel carrier fluxes depend on `phi`. The carrier Jacobian therefore acquires a **dense, phi-mediated tail** (measured: $\sim 13.5\%$ of the Jacobian Frobenius weight outside a 2-node band). In the near-insulating / collapsed-junction regime the dielectric-relaxation mode makes that coupling extreme, and the dense FD Newton goes **singular** (the existing ridge fallback + transient assists are the symptom-management). The full-thermal-velocity interface-plane-state block (`iface_state_v_th = 1e5`) adds a second stiff mode; today both share **one global dense Jacobian**, so they compound.

Gummel decoupling breaks the dense phi-coupling: solve the carrier-transport equations at *frozen* phi (block-tridiagonal, well-conditioned), then relax phi at *frozen* quasi-Fermi levels (the analytic, unconditionally well-posed dielectric step), and iterate. The two stiff modes are then handled by the operator each is well-conditioned for, instead of one ill-conditioned Newton.

3. The design

A Gummel **point solver** that replaces a single failed `solve_steady_state` point — used as an *opt-in fallback* before the transient fallback:

```
y = ensure_iface_block(y_seed, mat)
y = _qfl_poisson_relax(x, mat, y, V_app)          # Poisson half (EXISTS)
for it in range(max_outer):
```

```

# --- carrier half: continuity at FROZEN phi ---
phi = solve_poisson_once(y) # one solve, then held fixed
y = carrier_solve_frozen_phi(x, stack, mat, y, V_app, phi) # NEW (~80-150 lines)
y = underrelax_ln(y_old, y, omega) # ln-space, positivity-free
# --- interface-plane states: local per-interface subsystems ---
if mat.N_iface_state: y = solve_iface_states_local(x, mat, y, phi) # NEW (reuse solve_plane_densi
# --- Poisson half (EXISTS) ---
y = _qfl_poisson_relax(x, mat, y, V_app)
# --- convergence on the SAME coupled residual solve_steady_state uses ---
res = peak_density_scaled_residual(assemble_rhs(0, y, ...))
if res < tol: break
if res rose: omega = max(0.1, 0.5*omega) # adaptive damping (load-bearing)
if res > res_guard: raise SteadyStateError(...) # fail-loud, same contract
return y, res

```

Three pieces, two of which already exist:

Piece	Status	Source
Poisson half (nonlinear Poisson at frozen QFL, banded, damped)	EXISTS	<code>_qfl_poisson_relax</code> <code>steady_state.py:451-512</code>
Carrier half (continuity at frozen phi, block-tridiagonal banded solve)	NEW (~80-150 lines)	mirrors the Poisson half; <code>carrier_continuity_rhs</code> is already phi-pure
Interface-plane states as local per-interface subsystems	NEW (small)	template <code>solve_plane_densities</code> <code>interface_plane.py:385-433</code>

4. Why it is tractable (verified from source)

- **carrier_continuity_rhs is already phi-pure** — it takes phi as an argument and never solves Poisson (`continuity.py:101`, reads phi only at 132-139). A frozen-phi carrier residual is a direct call (zero edits) or a phi= kwarg on `assemble_rhs` (~5-10 lines wrapping `mol.py:1670-1695`).
- **The carrier Jacobian at frozen phi is block-tridiagonal** — SG flux is nearest-neighbour (`fe_operators.py:58-68`), divergence couples node i to i+1 only (`continuity.py:244`), recombination is node-local (`recombination.py:47`). So it solves with `scipy.linalg.solve_banded`, exactly as the Poisson half already does (`steady_state.py:496`).
- **The interface-plane state block is block-diagonal per interface** — each contributor writes only into its own base=4k slot (`interface_plane.py:213,473,577`; `mol.py:1885`), no inter-interface coupling. With phi + carriers frozen, each interface is a closed 4-unknown system solvable with the existing damped log-Newton (`solve_plane_densities`, `interface_plane.py:385-433`). The only bulk back-coupling is the eval-node drain (`mol.py:1907-1910`), folded as a lagged source into the next carrier sweep. (Today this block lives in the *global* dense Jacobian, `steady_state.py:298,344-353` — decoupling it is itself a win.)

5. Implementation plan (milestones)

M	Deliverable	Est.
M1	Frozen-phi carrier-continuity residual + banded carrier Jacobian (carrier_solve_frozen_phi); unit test vs assemble_rhs carrier rows at a converged state (must match).	1 d
M2	Gummel outer loop (_gummel_point_fallback) alternating M1 + _qfl_poisson_relax, adaptive ln-space under-relaxation, certify-or-raise on the peak-density residual. Wire as opt-in fallback before _transient_point_fallback at the two except SteadyStateError sites (run_jv_sweep_ss:602-610, solve_voc_ss:657-661); gate SOLARLAB_SS_GUMMEL (default OFF) + gummel_fallback=False kwarg.	1 d
M3	Interface-plane states as local per-interface subsystems inside the loop (reuse solve_plane_densities); validate on the deep-CBO points that currently gap (delta E_C <= -0.2).	1-1.5 d
M4	Deep-regime convergence hardening (damping schedule, omega floor, semi-implicit bulk drain) + the test surface in Section 8 + docs.	1.5-2 d

Core deliverable (M1-M4): converge the deep-CBO / collapsed-but-bracketed SS points -> ~4-6 days. This fills the CBO SS gaps (the user's ask).

6. Integration & safety (bit-identical default)

- The Gummel helper is reached **only after the coupled Newton already raised** `SteadyStateError` — every point that converges today never enters it. Even with the flag on, converging points stay **bit-identical**; only the currently-failing regime changes behaviour. Zero edits to the Newton loop.
- Default OFF via the established opt-in pattern (cf. `SOLARLAB_SS_JAC_REUSE`, `te_softness`, `iface_states`). On Gummel failure, **fall through** to `_transient_point_fallback` so existing behaviour is strictly preserved.
- Fail-loud contract kept: certify on the same peak-density residual scale (`steady_state.py:546-553`), raise on failure — no silent fallback.

7. CRITICAL caveat — Gummel does NOT flip the $N_d_{ETL} \leq 1e10$ xfail

The near-insulating xfail (`test_steady_state.py` low-doping gate) is a **physics/boundary-condition problem, not a convergence problem**. The voltage walk already *completes* there and finds **no $J=0$ crossing below 1.6 V** — both the SS driver and the transient flat-band result agree there is no crossing (the model's crossing sits near 1.29 V, above the detailed-balance ceiling). SCAPS reports $V_{oc} \sim 1.10$ V there only via its near-insulating **contact convention** (flat-band finite-S contact), which SolarLab does not apply by default.

Therefore: - Gummel gives **convergence** in the near-insulating regime (kills the dielectric-relaxation stiffness the coupled Newton chokes on) — necessary but **not sufficient**. - Flipping the xfail to xpass *additionally* requires a **near-insulating contact boundary condition** (the existing `flat_band_contacts` path is the candidate seam) so that a physical crossing exists to be found. - That combined effort is **~10-18 days with a go/no-go** after the first crossing attempt, and is **out of scope** for the core deliverable. The CBO gaps (Section 1.1) do *not* depend on it — they have a real crossing already.

8. Acceptance criteria / test surface

Core (M1-M4): 1. **Bit-identical default** — full unit + the SS parity suite (`test_ss_jv_matches_frozen_ion_transient`, `test_direct_voc_consistent_with_jv`) unchanged with the flag OFF, and unchanged for every point that converges today with the flag ON. 2. **M1 correctness** — `carrier_solve_frozen_phi` reproduces `assemble_rhs` carrier rows at a converged state to tolerance. 3. **Deep-CBO convergence (the headline)** — `run_jv_sweep_ss(gummel_fallback=True)` converges the $\Delta E_C \leq -0.2$ eV CBO points that currently gap, with FOM ($V_{oc}/J_{sc}/FF/PCE$) physical and the SS curve overlaying the transient there (CBO is band-offset-limited). 4. **Fail-loud preserved** — a genuinely unsolvable point still raises `SteadyStateError`.

Stretch (separate effort, Section 7): 5. $N_d_{ETL} \leq 1e10$ flips to xpass *only* in combination with the contact-BC change — tracked as its own task with a go/no-go.

9. Risks & mitigations

Risk	Mitigation
Gummel is only linearly convergent; stalls/oscillates at high injection (V_{oc} knee, large $n \cdot p$)	Adaptive In-space under-relaxation (ω start 1, halve on residual rise, floor ~ 0.1); fall through to the transient fallback on stall.
n, p form a 2×2 node block (recombination cross-term)	Iterate the node-local n - p coupling, or use a 2-field banded solve.
Interface bulk-drain ($v_{th}=1e5$) is large + lagged $\rightarrow$ oscillation	Semi-implicit / under-relaxed drain; keep interface-states local-solve inside the outer loop.
Frozen-phi supply exponential clamped at 30 $\rightarrow$ deep CBO can sit on the clamp and lose sensitivity	Re-project ϕ each outer sweep (not once); validate CBO sensitivity is retained.
Partial (non-converged) Gummel sweeps would shift validated FOM	Certify-or-raise on the coupled residual; never return a non-converged point.
OneDrive sync can overwrite an edited solver file before <code>git add</code>	Stage promptly (known repo gotcha).

10. Scope boundaries

- **In:** a Gummel point-solver fallback that converges collapsed-junction / deep-offset SS points where the coupled Newton fails (fills the CBO SS gaps); opt-in, default-off, bit-identical; interface-states handled as local subsystems.

- **Out:** the near-insulating contact-convention / $Nd_ETL \leq 1e10$ physics fix (separate task, Section 7); making Gummel the *default* SS path (it is a fallback only); any change to the coupled-Newton hot loop.